

Ali Hamed

Software Engineer | Python, C/C++, Automation, CI/CD, Linux, Cloud

ali.h.10j@gmail.com • +972-53-339-2826 • Haifa District, Israel • [GitHub](#) • [LinkedIn](#) • [Portfolio](#)
Languages: Arabic & Hebrew (native) • English (professional)

SUMMARY

Software Engineer with **2 years** of industry experience building production-grade automation, tooling, and test frameworks for complex distributed 4G/5G Open RAN systems at Parallel Wireless. Writes clean, maintainable code in Python and C/C++, designs reusable frameworks, integrates and improves CI/CD pipelines, and debugs across application, Linux, networking, and infrastructure layers. Comfortable across the full stack of modern engineering — containers, virtualization, cloud, and AI/LLM tooling — with strong CS fundamentals (OOP, data structures, algorithms). M.Sc. Information Systems & AI (GPA 93), B.Sc. Computer Science.

EXPERIENCE

Parallel Wireless — Software Automation Engineer	09.2024 – present · 2 years
- Built production automation, tooling, and services in Python and C/C++ for 4G/5G Open RAN systems. - Designed reusable frameworks with OOP, design patterns, and clean modular architecture. - Cut build-to-deploy from 8h to 2h (~75%) through Jenkins/Groovy CI/CD redesign. - Wrote and maintained C/C++ unit tests for low-level, multi-threaded components. - Shipped 3 LLM-powered tools to production (Python, LangChain, FAISS, REST/WebSocket APIs). - Hands-on AI: deliver solutions from development to production with measurable impact. - Automated regression and integration testing across 200–500 test cases (PyTest, Robot Framework). - Owned root-cause analysis across application, Linux, and network layers (Wireshark, PCAP, gdb). - Operated Docker, Kubernetes, KVM, VMware, and AWS (EC2, S3, Lambda) environments. - Provisioned a 20+ server lab: servers, switches, NICs, firewalls, BMC/IPMI, RF equipment. - Ran RF simulations with spectrum analyzers and channel emulators on 4G/5G networks. - Designed service interfaces, drove code reviews, and documented architecture in Agile/Scrum.	

EDUCATION

M.Sc., Information Systems & AI • University of Haifa	GPA 93 • Expected 2027
Coursework in Big Data Systems, Cybersecurity, Software Engineering, Machine Learning, and Deep Learning, with applied projects on real data and distributed systems.	
B.Sc., Computer Science • Ruppin Academic Center	GPA 83 • 2022 – 2025
Specialization in AI & Data Science. Strong foundation in OOP, data structures, algorithms, operating systems, computer architecture, and software development (C/C++, Python, Java).	

CERTIFICATIONS

NVIDIA Networking • Apr 2025	Certificate
NVIDIA AI Infrastructure and Operations Fundamentals • Sep 2025	Certificate
AWS Cloud Developing • Dec 2024	Certificate
AWS Cloud Practitioner Essentials • Nov 2024	Certificate
IBM Generative AI and LLMs • Apr 2025	Certificate

SKILLS

Languages	Python · C · C++ · Java · SQL · Bash · Shell · Groovy
CS Foundations	OOP · data structures · algorithms · multi-threading · memory management · operating systems · system design fundamentals
Frameworks & Testing	Pytest · Robot Framework · Selenium · Postman · Appium · unit / integration / system testing · REST APIs · WebSocket · REST API design · microservices concepts · concurrency · clean code
CI/CD & Tools	Jenkins · Groovy · Git · GitHub Actions · build & release · AI coding tools (Claude Code, Cursor)
Cloud, Containers & Virtualization	AWS (EC2, S3, Lambda) · Docker · Kubernetes · KVM · VMware
Systems & Networking	Linux (Ubuntu, CentOS) · embedded & real-time systems · TCP/UDP/IP · Ethernet · VLANs · Wireshark · PCAP · iPerf · tcpdump · gdb · BMC/IPMI
AI/ML	PyTorch · TensorFlow · HuggingFace · Scikit-learn · LangChain · LangGraph · RAG · vector search · LLM agents
RF & Telecom	spectrum analyzers · channel emulators · RF simulation & signal analysis · RF equipment · 4G/LTE · 5G NR · Open RAN · networking protocols · lab equipment (servers, switches, NICs, firewalls)
Applied AI Delivery	end-to-end AI delivery (prototype → production) · LLM agents · RAG · FAISS / vector search · LangChain · MCP / tool calling · prompt engineering · model serving (REST/FastAPI) · containerized deployment (Docker, Kubernetes) · GPU/CUDA · evaluation & error analysis · human-in-the-loop
Practices & Methodologies	Agile · Scrum · SDLC · system design · test-driven development · code review · Git workflows (branching, pull requests) · debugging · technical documentation · problem-solving · collaboration

VOLUNTEER

Volunteer Instructor • Hezkek LaAtid "Scientific Excellence", Fureidis High School	09.2025 – present
Teaching Computer Science fundamentals and programming to 40 students.	
Student Tutor • Ruppin Academic Center (Algebra, Python, Discrete Math)	01.2023 – 02.2024
"Perach" Program • Tutor & mentor for elementary school students	2022 – 2025